

Lab05

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Lab05

```
library(readr)
library(ggplot2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v stringr    1.5.1
v forcats    1.0.0      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
aeroportos <- read_csv("airports.csv") |>
  select(IATA_CODE, LONGITUDE, LATITUDE)
```

Rows: 322 Columns: 7

```
-- Column specification -----
Delimiter: ","
chr (5): IATA_CODE, AIRPORT, CITY, STATE, COUNTRY
dbl (2): LATITUDE, LONGITUDE
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

analisa_aeronave <- function(tail_number, arquivo) {

  callback_filtro <- DataFrameCallback$new(function(chunk, pos) {
    chunk |> filter(TAIL_NUMBER == tail_number)
  })

  voos_aeronave <- read_csv_chunked(
    arquivo,
    callback = callback_filtro,
    chunk_size = 100000
  )

  tabela_tidy <- voos_aeronave |>
    arrange(YEAR, MONTH, DAY, DEPARTURE_TIME) |>
    left_join(aeroportos, by = c("ORIGIN_AIRPORT" = "IATA_CODE")) |>
    rename(LONG_ORIGEM = LONGITUDE, LAT_ORIGEM = LATITUDE) |>
    left_join(aeroportos, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) |>
    rename(LONG_DESTINO = LONGITUDE, LAT_DESTINO = LATITUDE) |>
    mutate(VELOCIDADE_MEDIA = DISTANCE / (AIR_TIME / 60))

  min_long <- min(c(tabela_tidy$LONG_ORIGEM, tabela_tidy$LONG_DESTINO), na.rm = TRUE) - 3
  max_long <- max(c(tabela_tidy$LONG_ORIGEM, tabela_tidy$LONG_DESTINO), na.rm = TRUE) + 3
  min_lat <- min(c(tabela_tidy$LAT_ORIGEM, tabela_tidy$LAT_DESTINO), na.rm = TRUE) - 3
  max_lat <- max(c(tabela_tidy$LAT_ORIGEM, tabela_tidy$LAT_DESTINO), na.rm = TRUE) + 3

  mapa <- ggplot(tabela_tidy) +
    borders("world", fill = "gray96", colour = "gray85") +
    borders("state", colour = "white") +

    geom_segment(
      aes(
        x = LONG_ORIGEM, y = LAT_ORIGEM,
        xend = LONG_DESTINO, yend = LAT_DESTINO,
        color = VELOCIDADE_MEDIA,
        linewidth = VELOCIDADE_MEDIA
      ),
      alpha = 0.5,
      lineend = "round"
    ) +

```

```

    scale_linewidth_continuous(range = c(0.4, 1.2)) +
    scale_color_viridis_c(option = "magma") +
    guides(
      color = guide_legend(title = "Velocidade (mph)",
        linewidth = guide_legend(title = "Velocidade (mph)")
      ) +

    coord_quickmap(
      xlim = c(min_long, max_long),
      ylim = c(min_lat, max_lat)
    )
  return(list(
    tabela = tabela_tidy,
    grafico = mapa
  ))
}

```

```

analisa_aeronave("N407AS", "flights.csv")

```

```

-- Column specification -----
cols(
  .default = col_double(),
  AIRLINE = col_character(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  SCHEDULED_DEPARTURE = col_character(),
  DEPARTURE_TIME = col_character(),
  WHEELS_OFF = col_character(),
  WHEELS_ON = col_character(),
  SCHEDULED_ARRIVAL = col_character(),
  ARRIVAL_TIME = col_character(),
  CANCELLATION_REASON = col_character()
)
i Use `spec()` for the full column specifications.

$tabela
# A tibble: 1,234 x 36
  YEAR MONTH DAY DAY_OF_WEEK AIRLINE FLIGHT_NUMBER TAIL_NUMBER
  <dbl> <dbl> <dbl>      <dbl> <chr>          <dbl> <chr>

```

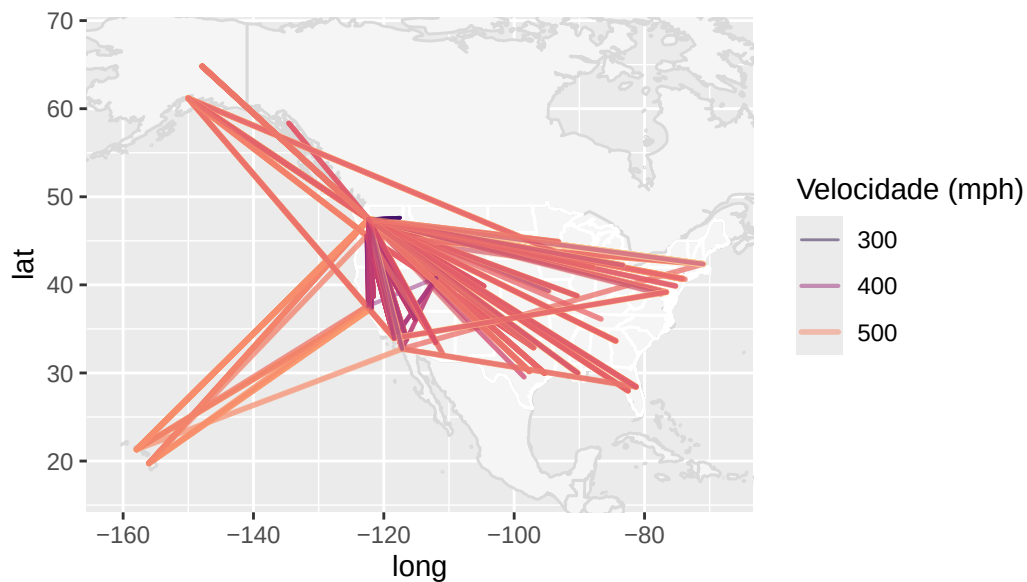
```

1  2015      1      1          4 AS          602 N407AS
2  2015      1      1          4 AS          617 N407AS
3  2015      1      1          4 AS           22 N407AS
4  2015      1      1          4 AS           21 N407AS
5  2015      1      1          4 AS          98 N407AS
6  2015      1      2          5 AS          81 N407AS
7  2015      1      2          5 AS          80 N407AS
8  2015      1      2          5 AS          14 N407AS
9  2015      1      3          6 AS          17 N407AS
10 2015      1      3          6 AS         452 N407AS
# i 1,224 more rows
# i 29 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
#   SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
#   TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
#   AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
#   SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
#   DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

```

\$grafico

Warning: Removed 113 rows containing missing values or values outside the scale range (`geom\_segment()`).



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-17 01:41:18 -03"
```